PROJECT PRESENTATION

FOOD DONATION MANAGEMENT PLATFORM

A Three-Role Platform for Food Donation Coordination and Delivery

DONATE • REQUEST • DELIVER

TEAM MEMBERS

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Connecting donors, NGOs and volunteers to reduce food waste.

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Project Overview

A COORDINATED PLATFORM FOR SURPLUS FOOD REDISTRIBUTION

FoodBridge is a web-based platform that coordinates surplus food donations and their delivery to people who need them.

The system connects three account roles—Donor, NGO and Volunteer—through a controlled workflow. Donors publish available food, NGOs request suitable quantities, and volunteers complete the delivery.

ONE SHARED WORKFLOW

DONOR

Publishes food details and manages incoming requests.

NGO

Discovers available donations and requests food.

VOLUNTEER

Accepts delivery assignments and updates progress.

Problem Statement

WHY A DIGITAL COORDINATION SYSTEM IS NEEDED

Food wastage

Edible surplus food may remain unused when there is no quick way to identify recipients.

Disconnected communication

Donors and NGOs may depend on calls or messages, making coordination difficult.

Delivery uncertainty

Without assigned volunteers and status updates, collection and delivery can be hard to track.

Core problem

There is a need for one structured platform that makes food availability visible to the right users and tracks each request from publication to delivery.

Project Objectives

WHAT FOODBRIDGE AIMS TO ACHIEVE

01

Digitise donations

Allow donors to publish food name, category, quantity, preparation and expiry details.

02

Enable NGO requests

Let NGOs browse available food and request a required quantity.

03

Control approvals

Allow only the donor who owns a donation to approve or reject its requests.

04

Coordinate delivery

Assign registered volunteers and track pickup through completion.

05

Improve visibility

Provide role-specific dashboards and account-scoped information.

06

Support accountability

Record delivery outcomes such as kilograms distributed and people served.

Existing System vs Proposed System

MOVING FROM INFORMAL COORDINATION TO A TRACKED WORKFLOW

EXISTING APPROACH

- Coordination through calls and messages
- Food details may be scattered across chats
- Request approval may not be recorded centrally
- Delivery progress can be difficult to monitor
- Limited role-based visibility

PROPOSED: FOOD DONATION PLATFORM

- Centralised donation listing
- NGO requests linked to specific donations
- Donor-controlled approval or rejection
- Volunteer assignment and delivery status tracking
- Separate views and counts for each account role

User Roles and Responsibilities

EXACTLY THREE ACCOUNT ROLES — NO ADMIN ACCOUNT IN THE INTENDED SYSTEM

DONOR

- Register and sign in
- Publish food donations
- View own donations
- Review requests for own food
- Approve or reject requests
- Assign a registered volunteer

NGO

- Register and sign in
- Browse all available donations
- Search/filter food listings
- Request a required quantity
- Track own requests and outcomes

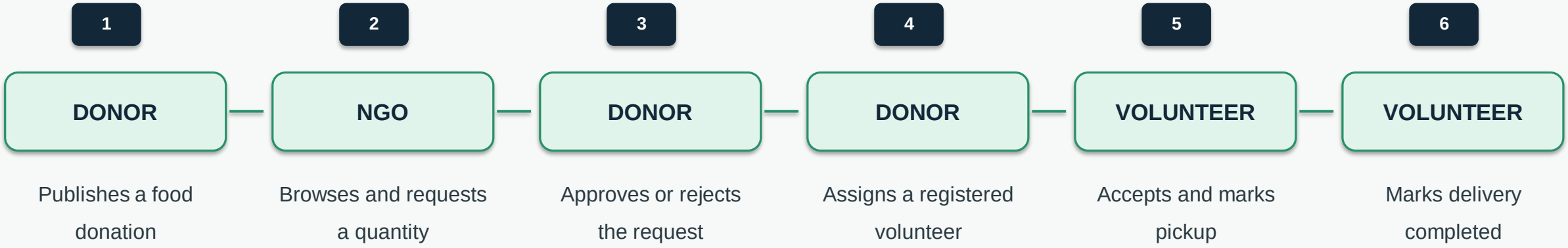
VOLUNTEER

- Register and sign in
- View assigned deliveries only
- Accept an assignment
- Mark food as picked up
- Mark delivery as completed

Data visibility is role-scoped: users see only the records relevant to their role and account.

End-to-End System Workflow

FROM SURPLUS FOOD PUBLICATION TO COMPLETED DISTRIBUTION



Status progression

Request: PENDING → APPROVED / REJECTED | Assignment: PENDING → ACCEPTED → PICKED_UP → DELIVERED

Major System Modules

FUNCTIONAL AREAS IMPLEMENTED IN THE APPLICATION

Authentication

Registration and login for the three roles; JWT-based authentication.

Donation Management

Create and view food donations with quantity, category, preparation and expiry details.

Food Request Management

NGOs submit requests; donors approve or reject requests tied to their donations.

Volunteer Assignment

Donors assign registered volunteers to approved requests.

Delivery Tracking

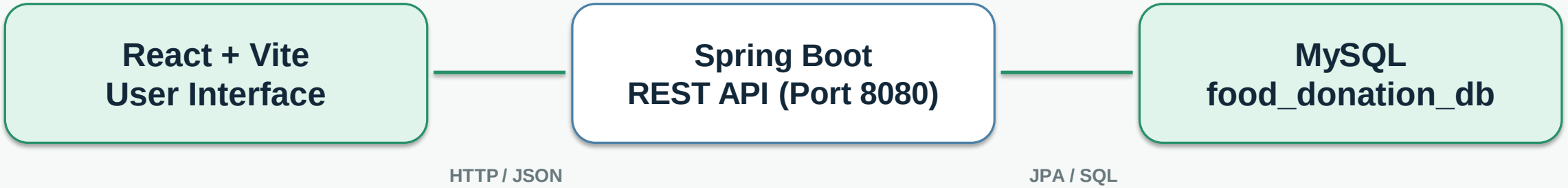
Volunteers update assignment status as accepted, picked up and delivered.

Role-Based Dashboard

Shows role-specific summary counts and distribution impact.

System Architecture

FRONTEND, REST API, APPLICATION SERVICES AND RELATIONAL DATABASE





The frontend sends authenticated API requests. Spring Boot applies role and ownership checks, repositories access MySQL through Spring Data JPA, and responses are returned to the interface.

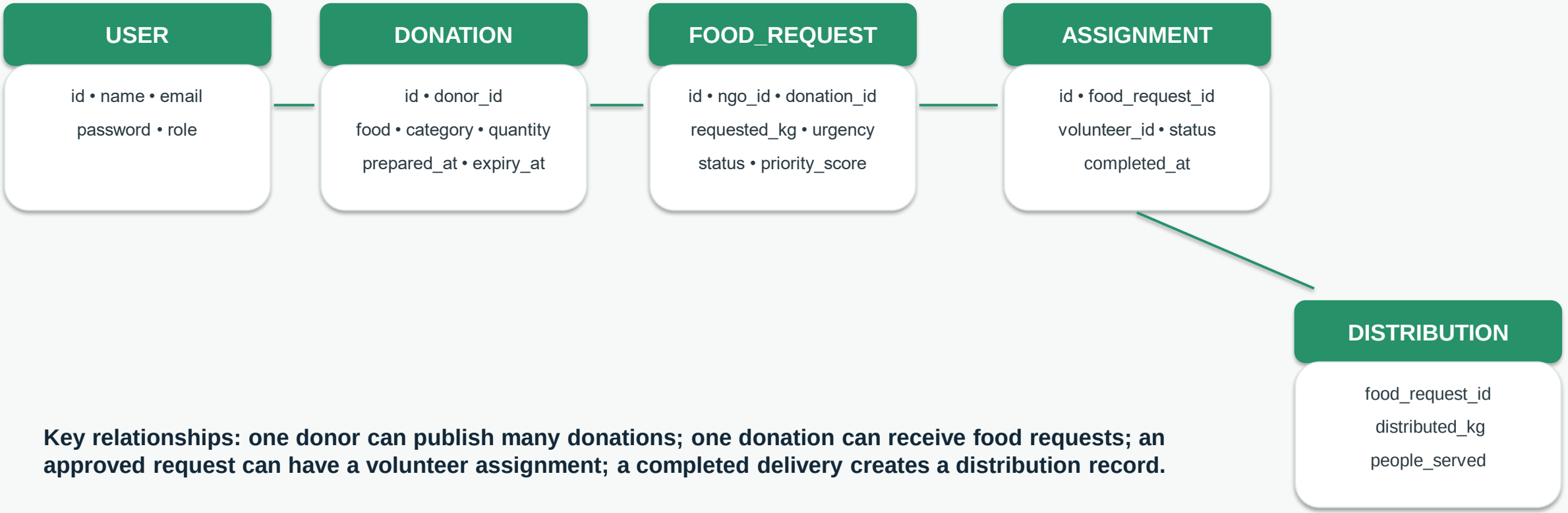
Technology Stack

TECHNOLOGIES IDENTIFIED IN THE PROJECT SOURCE

Frontend	**React.js • Vite • JavaScript • CSS**	Builds the interactive role-based web interface.
Backend	**Java 17 • Spring Boot 3.4.5**	Provides REST endpoints and application logic.
Persistence	**Spring Data JPA • Hibernate**	Maps Java entities to relational database records.
Database	**MySQL 8**	Stores users, donations, requests, assignments and distributions.
Security	**Spring Security • JWT**	Protects API access using authenticated tokens and role checks.
Build / Runtime	**Maven • Node.js 18+**	Builds and runs the backend and frontend applications.

Database Design / Main Entities

CORE RECORDS REPRESENTED IN THE BACKEND MODEL



Key relationships: one donor can publish many donations; one donation can receive food requests; an approved request can have a volunteer assignment; a completed delivery creates a distribution record.

Security and Data Access

PROTECTING ACCOUNTS AND KEEPING INFORMATION RELEVANT

JWT AUTHENTICATION

Users sign in and access protected API endpoints using a JSON Web Token.

ROLE CHECKS

Actions are restricted by role: NGOs request, donors decide and assign, volunteers update delivery.

OWNERSHIP CHECKS

Donors can decide requests only for their own donations; volunteers can update only their own assignments.

Role-scoped visibility

- Donor: sees donations they published and requests made against those donations.
- NGO: sees available donations and only its own food requests.
- Volunteer: sees only delivery assignments assigned to that account.

Dashboard and Delivery Tracking

OPERATIONAL VISIBILITY AND IMPACT INDICATORS

Role-specific summary metrics

DONOR VIEW

Total donations • available packages

Pending requests • people served

NGO VIEW

Available donations • own pending requests

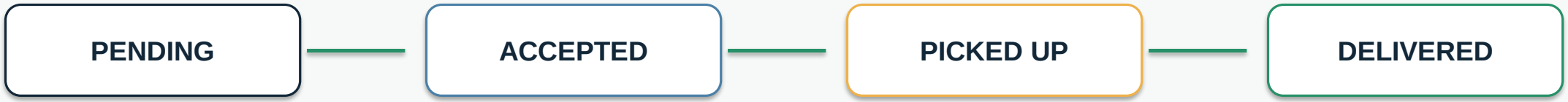
Completed distributions • food received

VOLUNTEER VIEW

Assigned deliveries • pending tasks

Completed deliveries • people served

Delivery status lifecycle



Dashboard data is refreshed after actions, when the browser tab regains focus, and periodically while signed in.

Implementation and How to Run

LOCAL SETUP BASED ON THE PROJECT README

BACKEND — JAVA / MAVEN

1. Install Java 17+, Maven 3.9+ and MySQL 8.

2. Create the database:
   CREATE DATABASE food_donation_db;

3. Configure DB username/password in:
   backend/src/main/resources/application.properties

4. Open terminal in backend and run:
   mvn spring-boot:run

Backend: http://localhost:8080

FRONTEND — REACT / VITE

1. Install Node.js 18+.

2. Open a second terminal in frontend.

3. Install dependencies:
   npm install

4. Start the development server:
   npm run dev

5. Open the local URL shown by Vite, usually:
   http://localhost:5173

Important Validation Scenarios

SUGGESTED DEMONSTRATION AND TESTING CHECKLIST

- Register three separate accounts: one Donor, one NGO and one Volunteer.
- Sign in as Donor and publish a donation with a future expiry time.
- Sign in as NGO, locate the available donation and request a valid quantity.
- Return to the owning Donor account and approve or reject the request.
- Assign a registered Volunteer to an approved request.
- Sign in as that Volunteer: accept, mark pickup, then mark delivery complete.
- Verify that the NGO and Donor can see the updated request and delivery outcome.

Note: the project README states that full end-to-end testing against the local Windows/MySQL setup is still required.

Advantages and Current Limitations

WHAT THE PROJECT DELIVERS AND WHAT SHOULD BE VALIDATED

ADVANTAGES

- Centralised food donation workflow
- Clear responsibilities for three roles
- Ownership-based request decisions
- Volunteer delivery status tracking
- Impact metrics for food and people served
- Role-scoped data visibility

CURRENT LIMITATIONS / VALIDATION

- Requires internet/local network and configured services
- Depends on MySQL availability and correct credentials
- No live GPS route tracking is indicated in the source
- No payment or food-quality verification module is indicated
- Full end-to-end local testing is still required

Future Enhancements

POSSIBLE IMPROVEMENTS BEYOND THE CURRENT IMPLEMENTATION

Live location tracking

Show volunteer location and delivery progress on a map.

Notifications

Send email, SMS or in-app alerts for new requests and status changes.

Food safety checks

Add freshness checks, storage guidance and verification records.

Analytics

Provide trends for donated kilograms, response time and distribution impact.

Mobile experience

Create a responsive mobile app or progressive web application.

Smart matching

Recommend donations to NGOs based on location, quantity and urgency.

CONCLUSION

A connected path from surplus food to people in need.

FoodBridge brings donors, NGOs and volunteers into one coordinated process. By structuring donations, approvals, assignments and delivery updates, the system aims to reduce coordination gaps and make food redistribution more accountable.

THANK YOU

Questions & Discussion